

Response to the Editor and Reviewers

Submission ID: 1e58c388-d7ec-4084-b792-df20a48e3e75. Locations refer to section titles and figure/table labels.

We thank the editor and reviewers for their careful reading. We have clarified the horizon definitions, explained their practical use, and revised the presentation of the four-domain results. We also checked the stored evidence behind the numerical summaries and the alternative-baseline analysis. The corrections described below distinguish changes to the reporting from changes to experiments.

Additional experiments carried out in this revision. No new model class, hyperparameter, split or evaluation protocol was introduced. We carried out the following analyses, all on the same data and protocol as the paper:

1. *Wind ARIMA check with a true h -step forecast.* The earlier check requested a one-step forecast at every horizon. We reran it with the h -step forecast; the wind descriptors are unchanged (Section Wind Domain Results).
2. *ARIMA order diagnostic.* An AIC search ($p, q \leq 5, d = 0$) on the training windows of 20 evenly spaced evaluated origins per domain (Appendix A.2). The reported ARIMA check was not refitted with these orders.
3. *RMSE recomputation.* The four LightGBM experiments were re-executed unchanged while also recording RMSE. Their MAE reproduces the reported MAE exactly, and Appendix A.1 now uses these per-origin errors.
4. *Seasonal-persistence sensitivity.* Re-executed on the current inputs; all eight reported descriptor tuples are reproduced exactly (Table 4).

Editor

We thank the Editor for handling the manuscript. Besides the changes requested by the reviewers, we made the following corrections to the reporting.

We corrected a mismatch between the PM_{2.5} text and its plotted LightGBM curve. That curve gives $H^*(\text{strict}) = 22$ and [27, 48]. The former value of 13 and [36, 48] came from a separate moving-average result, not the reported LightGBM curve. We corrected the text, caption, and Table 3 without changing the forecasts or the figure data.

In “Models”, the daily-load description now follows the producer: expanding training, evaluation beginning after 365 observations, and a weekly index-phase feature rather than a timestamp-derived day-of-week label. We removed claims that the observed profiles establish intrinsic domain limits or robustness across forecasting models. “Limitations” now separates positive average skill from evidence of operational benefit and makes the baseline, metric, support, model, and horizon caps explicit. The discussion of dynamic horizon selection remains a potential use, not a validated controller.

The archived four-domain seasonal-baseline analysis is reported in “Alternative-baseline sensitivity” and Table 4. Its stored comparisons were checked on exact common support, and the analysis was re-executed on the current inputs, reproducing all reported values exactly. The metric-sensitivity appendix (Appendix A.1) was recomputed from per-origin squared errors of the same LightGBM forecasts. The rerun reproduces the reported MAE exactly in all four domains, so the MAE and RMSE descriptors share forecasts and support. This corrected the PM_{2.5} RMSE row (48/23, [26, 48]) that had been inherited from a different, moving-average experiment; the other three rows are unchanged (wind 48/48, traffic 72/63, load 3/3).

Reviewer 1

We follow the seven points of the report in their original order.

Comment 1

In section 3.3.1. Change “.” to “:” before the explanation and mathematical description of the relaxed and strict horizons.

Response: We replaced the full stop with a colon after both labels in “Operational Predictability Horizon” (Section 3.1; the reviewed PDF numbers the "Operational definitions" paragraph as Section 3.1.1).

Comment 2

In section 3.3.1 in equation (5) include the condition of including the non-positive intermediate skills within the horizon as the definition using equation (5) does not represent what $H^(\text{relax})$ is extracting from the curve.*

Response: We now define

$$\mathcal{H}^+ = \{h \in \mathcal{H} : \text{Skill}(h) > 0\}$$

and

$$H^*(\text{relax}) = \max(\mathcal{H}^+ \cup \{0\})$$

in Section 3.1. Equation (5) also defines the relaxed span $\mathcal{R}$ and the gap set $\mathcal{G} = \mathcal{R} \setminus \mathcal{H}^+$, whose elements have non-positive skill. This set may be non-empty.

Comment 3

In this definition of relax H^ reiterate for clarity purposes here that it considers all the time steps before as it skill score once the upper bound of the interval is positive as seen in the PM2.5 and the traffic domain results.*

Response: The text explicitly notes that intermediate horizons with non-positive skill may fall within the relaxed reach, but are not themselves interpreted as skilful. In the PM_{2.5} and traffic results we now also state how much of the relaxed span has non-positive skill: 19 of the 48 PM_{2.5} horizons and 36 of the 72 traffic horizons. We verified this behaviour using executable unit tests covering intermediate skill deficits and delayed onset.

Comment 4

Keep units throughout in Table 2.

Response: Table 2 now specifies $\text{PM}_{2.5}$ in $\mu\text{g m}^{-3}$, wind speed in m s^{-1} , and traffic speed in mph. For electric load, we clarify that the stored experimental target is the sum of 15-minute kW readings and that division by four expresses the aggregate daily demand in kWh, following the source metadata and the aggregation script.

Comment 5

Figures 1-3 are not referenced in text.

Response: We have added explicit textual references to the conceptual H^* schematic (Fig. 1), the rolling-origin validation protocol (Fig. 2), and the $\text{PM}_{2.5}$ skill curve (Fig. 3).

Comment 6

An evaluation of other baseline choices is warranted, other than persistence. For some datasets you used $\text{ARIMA}(2,0,0)$ which is essentially an $\text{AR}(2)$ model why did you choose this parameter? Did you do an ACF plot for this data?

Response: *Alternative baselines.* We include the archived seasonal-persistence analysis, using a fixed 24-hour period for the hourly domains and a 7-day period for daily load. For each domain, we verified equality of forecast origin, target timestamp, horizon, and y_{true} between the stored LightGBM, persistence, and seasonal-persistence records. Pairwise alignment loses no records. The analysis was re-executed on the current inputs and reproduces all eight reported descriptor tuples exactly.

For $\text{PM}_{2.5}$, the persistence comparison on this exact common support yields $H^*(\text{strict}) = 22$ with longest contiguous positive interval $[27, 48]$, whereas seasonal persistence yields $H^*(\text{strict}) = 24$ with $[25, 48]$.

The primary LightGBM curve also gives $H^*(\text{strict}) = 22$ and $[27, 48]$, after the reporting correction explained above. The previous 13-hour value is not explained as a support effect. Equal descriptors do not imply equal samples or MAEs: the seasonal-baseline comparison uses only its matched-support rows.

Under the same matched-support analysis, load remains at $H^*(\text{strict}) = 1$, wind changes from 48 to 30, and traffic changes from 7 to 14. $H^*(\text{relax})$ remains unchanged across all four domains. We note that the hourly endpoints are exact multiples of the seasonal period, where seasonal and standard persistence coincide; the unchanged relaxed endpoint is therefore partly structural and is not interpreted as evidence of baseline invariance.

ARIMA order and ACF. We agree that the original manuscript did not document this choice. The order $(2, 0, 0)$ was fixed a priori as a low-order autoregression; an audit of our repository found no record that it was selected via ACF/PACF, information criteria, grid search, auto-ARIMA or validation splits, and we do not claim otherwise. The ARIMA check is now described in Section Models as a model-substitution illustration, not a tuned competitor.

To answer the question directly, we examined the data. On the training windows of 20 evenly spaced evaluated origins per domain, an AIC search ($p, q \leq 5$, $d = 0$, with constant) selects different orders at different origins (10 distinct best orders in wind, 8 in traffic), and $(2, 0, 0)$ lies within $\Delta\text{AIC} \leq 2$ of the optimum in 10 of 20 wind windows but only 4 of 20 traffic windows (Appendix A.2). We report this as a limitation of the fixed order, not as a justification. The ACF and PACF of the last training window of each domain are shown in a new figure in

Appendix A.2; they were inspected after the order was fixed. The PACF is dominated by lag 1, and the traffic ACF has a daily peak at lag 24 that a non-seasonal AR(2) cannot represent.

The audit also found that the historical wind check requested a one-step forecast at every horizon instead of the stated h -step forecast. We reran it with a true h -step forecast, otherwise identical: the wind descriptors are unchanged ($H^*(\text{relax}) = H^*(\text{strict}) = 48$, $[1, 48]$, no sign changes) while skill amplitude differs. Traffic used an h -step forecast and its numbers are unchanged ($H^*(\text{strict}) = 6$, $[38, 43]$, 13 sign changes).

The separate seasonal-baseline analysis tests sensitivity to the reference baseline, not robustness across forecasting models, and is not used to justify the ARIMA order.

Comment 7

State or reiterate the practical usefulness and use cases of each skill- relaxed and strict. Also, examine what can be done with these patterns that regular fixed horizon accuracy summaries cannot do. For example, can you have the persistence baseline and skill score – relaxes and strict– be used to determine the various h steps to predict in a rolling window approach for a dynamical forecast of the time series? Or would this create a feedback loop and cause overfitting. Explore and if feasible add to the paper to make it more substantial and unique.

Response: A new “Operational interpretation and actionability” subsection (Section 3.2) defines $H^*(\text{relax})$ as the outer horizon at which positive baseline-relative skill is observed, and $H^*(\text{strict})$, with its interval location, as the longest contiguous positive interval.

We explain that a few fixed-horizon errors do not locate positive-skill intervals or show the gaps between them. The PM_{2.5} and traffic examples illustrate why interval locations are needed in addition to lengths. We also state that positive average skill alone does not establish deployment value: its magnitude, uncertainty, decision loss, and cost matter.

Dynamic use. We discuss this possibility in “Discussion”. In an operational setting, H^* could inform a decision policy estimated on historical validation data, frozen for prospective deployment, and subsequently updated using only realised historical errors.

We explicitly caution that estimating H^* , selecting horizons, and evaluating performance on the same errors introduces adaptive selection bias. A dynamic controller would require prospective, prequential, or nested validation, and is not claimed to be validated here.

Reviewer 2

Comment 1

The paper needs a more explicit explanation of the actionability of its findings.

Response: We added the dedicated operational interpretation subsection noted above (Section 3.2), distinguishing the outer positive reach from the longest contiguous positive interval and restricting dynamic deployment claims to future, separately validated policies.

Comment 2

Figures 1 and 2 are not referenced in the text.

Response: Both figures are now explicitly introduced and discussed in the main text (Figs. 1–2).

Comment 3

In fig 1, parts of the line obscure some of the text.

Response: The revised Figure 1 uses dedicated callout boxes and separated label placements. This is a layout correction, not a change to the horizon definitions or empirical results.

Comment 4

In fig 2, the text does not fit inside the boxes.

Response: The revised Figure 2 has larger boxes and wrapped text.

Comment 5

If possible please evaluate alternative baseline options beyond persistence.

Response: We report the archived seasonal-persistence analysis detailed in our response to Reviewer 1 (Comment 6), enforcing exact common support and reporting all outcomes regardless of direction. Table 4 reports all four domains across both baselines, documenting common-origin sample counts, $H^*(\text{relax})$, $H^*(\text{strict})$, and interval locations.

Comment 6

“Data Availability” It is too much of a burden to have to go to 4 or 5 sources. Please cache all the data in a site you control and point to it (stop press, it seems you do that on the code page. Just make this clear in your paper).

Response: We deposited the exact input files of all four experiments in the project repository, with SHA-256 checksums and provenance notes, under the fixed tag `data-cache-v1`: https://github.com/EduIntellect/paper2H/tree/data-cache-v1/release/data_cache_four_domain. The “Data Availability” section now points to this location. $\text{PM}_{2.5}$ and electric load are redistributed under CC BY 4.0 and wind under NREL’s open-data terms. The traffic file is a single-sensor hourly aggregate of METR-LA; its underlying measurements come from Caltrans PeMS, and we state in the repository that we did not obtain separate confirmation of redistribution terms and will remove the file on request.

Comment 7

For certain datasets, $\text{ARIMA}(2,0,0)$ —effectively an $\text{AR}(2)$ model—was applied. Could you explain the rationale for selecting these parameters?

Response: As noted in our response to Reviewer 1 (Comment 6), the order $(2, 0, 0)$ was fixed a priori as a low-order autoregression and was not selected with ACF/PACF or information criteria. We now say so in Section Models, report an a posteriori AIC-based diagnostic in Appendix A.2 (the best order varies across origins, and $(2, 0, 0)$ is further from the optimum in traffic than in wind), and present the check as an illustration of descriptor sensitivity, not a tuned model. The wind check was rerun with a true h -step forecast; its descriptors are unchanged.